Laboratory 1: Waveguides

Hugo Beltrán Sanz, Group B1

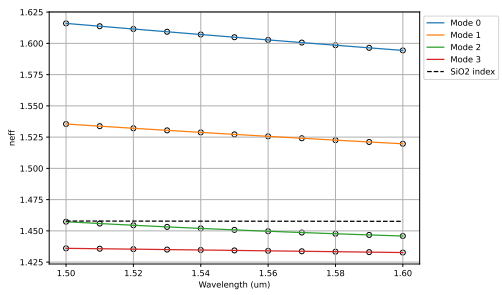
1 Effective index of a waveguide

The guided solutions of a dielectric waveguide are characterized by their propagation constant β. The corresponding effective index is defined as

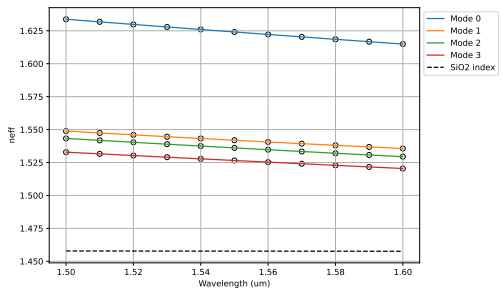
$$n_{\text{eff}} = \frac{\beta}{k_0}.$$

For a bound mode in a dielectric waveguide, a practical screening condition is $n_{\text{clad}} < \Re\{n_{\text{eff}}\} < n_{\max}$, where $n_{\max}$ is the largest material index in the structure. This check is useful but not sufficient on its own; field profiles, leakage (or $\Im\{n_{\text{eff}}\}$), and solver artifacts should also be considered.

In the simulations, the mode solver returns complex n_{eff} values for the computed modes. The TE fraction is also provided and is useful to identify polarization character (TE-like or TM-like), but not as a stand-alone proof of guidedness. For example, Mode 2 (the third computed mode) is TE-like in polarization content, yet its effective index is close to the cladding index and must be interpreted cautiously regarding confinement.



(a) Deep SiNx waveguide



(b) Shallow SiNx rib waveguide

Figure 1: Effective index versus wavelength for deep and shallow SiNx waveguides.

2 Wavelength behavior

The effective index varies with wavelength over the analyzed spectral window. For the SiNx platform, both a deep strip waveguide and a shallow rib waveguide were simulated in the 1.50–1.60 μm range. Figure 1 compares both structures.

Relative to the deep waveguide, adding a 150 nm slab increases n_{eff} for the computed modes and shifts higher-order modes toward a more guided regime. This trend is consistent with the added high-index slab, which modifies the modal distribution and increases the effective index of the supported solutions.

3 Width dependence

The deep SiNx waveguide was then evaluated at $\lambda = 1.55\,\mu$m while sweeping width from 0.6 to 1.6 μm (Fig. 2). Using the solver indexing (Mode 0, Mode 1, ...), Mode 0 remains TE-like and above the cladding-index threshold across the simulated range, while Mode 1 is predominantly TM-like and also remains above that threshold. Mode 2 approaches and then exceeds the cladding-index threshold at larger widths, whereas Mode 3 stays near or below cutoff in this range.

Therefore, around 1.2 μm width, the modal behavior is largely set by Mode 0 and Mode 1, while additional higher-order solutions become relevant as the width increases.

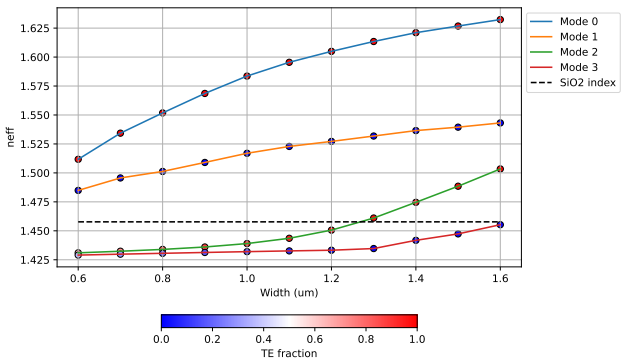


Figure 2: Effective index versus width for the deep SiNx waveguide at 1.55 μm.

4 Waveguide compact model

A second-order compact model was fitted to the fundamental-mode dispersion:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2, \quad \lambda_0 = 1.55\,\mu\text{m}.$$

From this fit:

$$n_{\text{eff}}(\lambda_0) = a_0, \qquad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}\Big|_{\lambda_0} = a_0 - \lambda_0 a_1,$$

$$D = -\frac{\lambda}{c}\frac{d^2 n_{\text{eff}}}{d\lambda^2}\Big|_{\lambda_0} = -\frac{2\lambda_0 a_2}{c}.$$

The fit quality is shown in Fig. 3, and extracted parameters are listed in Table 1.

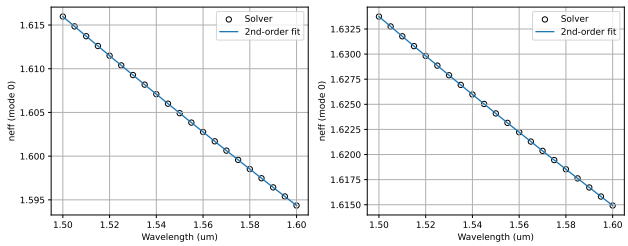


Figure 3: Second-order compact-model fit of $n_{\text{eff}}(\lambda)$ for deep and shallow SiNx waveguides.

Table 1: Compact-model parameters at $\lambda_0 = 1.55\,\mu$m (fundamental mode).

Waveguide	$n_{\text{eff}}(\lambda_0)$	n_g	D [ps/(nm km)]
Deep SiNx	1.6049	1.9400	-0.0010
Shallow SiNx	1.6241	1.9158	-0.0010

The shallow waveguide shows slightly larger $n_{\text{eff}}(\lambda_0)$, slightly smaller n_g, and comparable fitted D. Here D should be interpreted as a local dispersion parameter extracted from the quadratic fit at λ_0. Since $|D|$ is very small in these results, unit/conversion consistency should be checked when comparing with other tools or references.

5 Bend radius versus loss

For the deep SiNx waveguide, the bend-overlap trend is shown in Fig. 4. If η is the mode-overlap factor ($0 < \eta \leq 1$), then $10\log_{10}\eta$ is negative, while the corresponding mismatch loss is $-10\log_{10}\eta$ in dB. In this report, total bend loss is computed as mismatch loss plus bend propagation/radiation loss over a 90° arc.

The decomposition of losses is shown in Fig. 5. With the criterion $L_{\text{total}} < 0.1\,\text{dB}$ per 90° bend, the first acceptable radius in the simulated set is $31\,\mu$m, which is taken as the safe bend radius for the deep SiNx case.

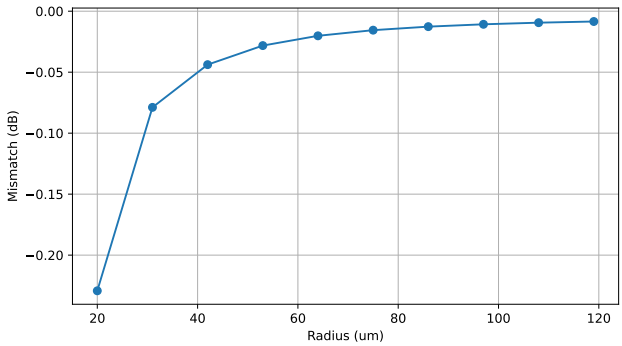


Figure 4: Bend mode-overlap metric $10\log_{10}(\eta)$ versus radius (deep SiNx).

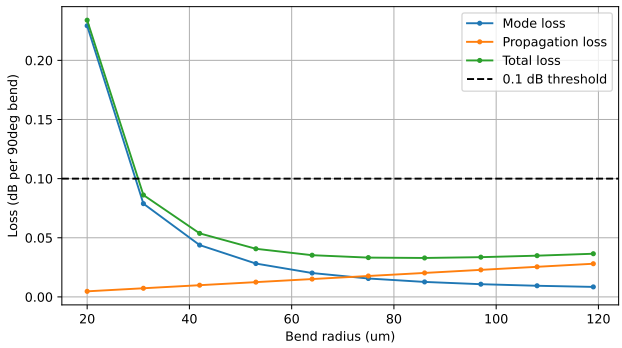


Figure 5: Mode-mismatch, bend propagation/radiation, and total loss per 90° bend for deep SiNx.

6 SOI technology benchmark

The same workflow was repeated for an SOI strip waveguide (220 nm height, 500 nm width). Figures 6 and 7 show wavelength and width sweeps, respectively.

Compared with SiNx, SOI exhibits larger effective indices due to stronger index contrast (Si/SiO_2) and shows earlier appearance of higher-order modes as width increases within the simulated sweep.

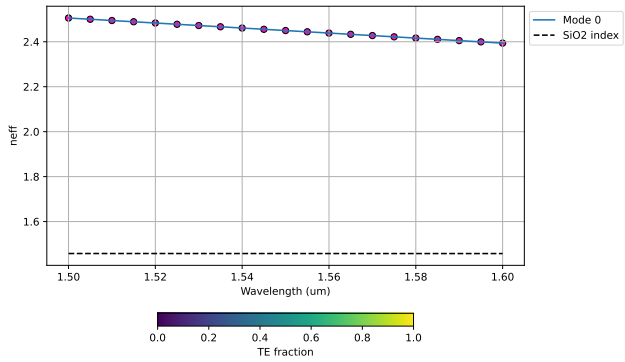


Figure 6: SOI strip effective index versus wavelength (220×500 nm).

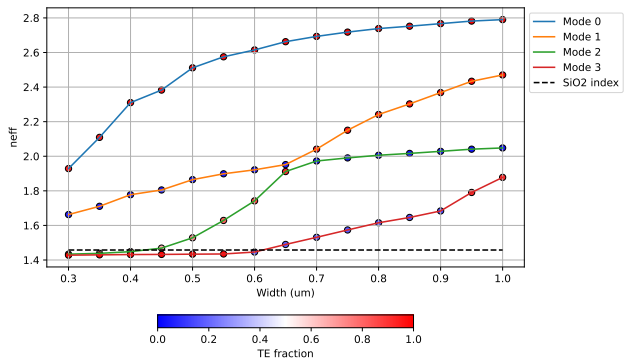


Figure 7: SOI strip effective index versus width at $1.55\,\mu$m.

The SOI bend-loss sweep is presented in Fig. 8. Using the same threshold (< 0.1 dB per 90° bend), the minimum simulated radius ($5\,\mu$m) already satisfies the criterion.

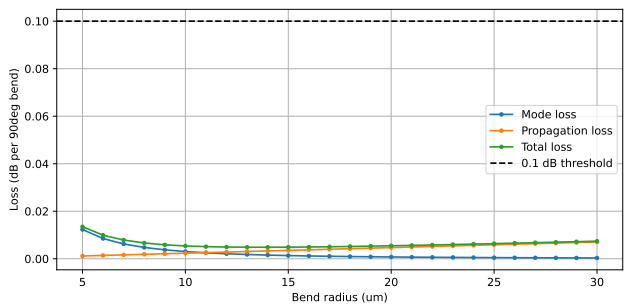


Figure 8: SOI bend-loss components versus bend radius.

Table 2: Comparison between deep SiNx and SOI (fundamental mode at $\lambda_0 = 1.55\,\mu$m).

Technology	$n_{\text{eff}}(\lambda_0)$	n_g	D [ps/(nm km)]	R_{safe} [μm]
Deep SiNx	1.6049	1.9400	-0.0010	31
SOI	2.4499	4.1814	+0.0004	5

Overall, the SOI configuration considered here enables much tighter bends and larger effective indices than the SiNx deep waveguide. The same high-index contrast that improves compactness also promotes earlier multimode behavior as the transverse dimensions are increased.